AI Application · Mocap3D (2D video → 3D) · Sonny · 10 November 2026

Follow-up experiments from the paper review

We took one testable idea from each paper in *Papers on the lifting models* and *Lifting Models for Our Project*, wrote the success criteria before running, and tested it on KERAAL Group 3 (against Vicon) and on REHAB24-6. No new training was needed: every test uses the current lifter and its three leave-one-participant-out models.

Main findings

1. **Temporal context is what makes our lifter smooth** (VideoPose3D idea). Removing the 81-frame context at test time nearly tripled the velocity error (7.0 → 19.5 mm/frame) and added 6.6 mm of position error. Our skeleton is also smoother than Kinect's (velocity error 7.0 vs 11.5 mm/frame).
2. **Depth is our weak axis, but not Kinect's** (REHAB24-6 idea). Our largest error is along the camera's depth axis (38 mm, against 23–26 mm across the image). Kinect, a depth sensor, has its *smallest* error in depth (23 mm). Torso rotation (RTK) has the largest depth error (44 mm), which matches the earlier RTK problem.
3. **A DTW comparison with correct repetitions detects some errors, but not all** (3D feedback idea). With no training at all, our 3D gives AUC 0.73 on CTK and 0.70 on RTK, about the same as Kinect. ELK is at chance (≈0.52) for every source, including Vicon. Naming the joint that is wrong agrees with Vicon only 22–35 % of the time (chance 12.5 %). This is not yet good enough for joint-level feedback.

1. Which paper led to which experiment

Paper (from the review)	Idea we tested	Experiment
Research 2 — VideoPose3D (Pavllo et al., 2019)	Many frames make 3D smoother and more accurate than one frame. The paper reports about 76 % lower velocity error (MPJVE) than single-frame lifting.	**E1**: temporal context and smoothness
Research 3 — REHAB24-6	Depth is the main difficulty for 3D from video. Models should be tested across people, cameras and exercises.	**E2**: error by axis, camera and exercise
Research 4 — 3D Pose-Based Feedback (EC3D)	Compare a movement with correct movements using DTW, then find the body part that is wrong. The EC3D pipeline also smooths skeletons with a Hamming filter.	**E3**: DTW error detection and localisation; the Hamming filter is used in E1
Research 1 — UCO Physical Rehabilitation	Five camera views, 16 exercises (seated, standing, lying).	Not tested: the dataset is available on request only. See section 6.

2. Setup (the same for every experiment)

- **Model**: our VideoPose3D-style TCN lifter (81 frames, root normalisation, YOLO11x-pose 2D input). It was trained on ASPset-510, KERAAL (G3 with Vicon; 1B, 2A and 2B with Kinect targets) and REHAB24-6 (persons 1 and 2 held out). This is the 8 November model (`km/kin_w_r24`).
- **KERAAL G3**: three participants and 540 recordings (CTK, ELK, RTK; correct and acted errors). Each participant is tested with the model that never saw them (leave one participant out). Ground truth is Vicon in the Kinect camera frame, relative to the pelvis.
- **Kinect**: the Kinect v2 skeleton of the same recordings, in the same frame.
- **REHAB24-6**: the 36 test views (persons 1 and 2, cameras c17 and c18), using all three fold models.
- **Pre-registration**: predictions and measures were written at the top of `paper_followup/followup.py` before the first run.

3. E1 — Temporal context and smoothness (VideoPose3D)

Question: does our lifter gain from looking at 81 frames, as VideoPose3D does, and is its output smooth enough for feedback?

Method: on each held-out G3 participant we measured:

- **MPJPE**: position error.
- **MPJVE**: velocity error, i.e. how wrong the movement between two frames is.
- **Jitter**: mean acceleration, i.e. shakiness. Vicon's own jitter is 4.0.

"No context" feeds the same model a window filled with one repeated frame. This is a test-time ablation, not a trained single-frame model, so it overstates the single-frame penalty somewhat. The "+ Hamming" rows apply a 0.3 s Hamming filter, as in the EC3D pipeline.

Source (mean of 3 participants)	MPJPE (mm)	MPJVE (mm/frame)	Jitter (mm/frame^2)
Ours (81-frame context)	59.6	**7.0**	5.1
Ours, no context	66.2	19.5	28.8
Ours + Hamming	59.6	5.7	1.2
Ours, no context + Hamming	63.9	7.5	2.9
Kinect	52.2	11.5	14.8
Kinect + Hamming	51.1	6.3	2.3

Prediction (written before running)	Result
Removing context raises MPJVE by at least 50 %	Supported +180 % (7.0 → 19.5), and MPJPE +6.6 mm
Kinect has higher MPJVE than ours	Supported 11.5 vs 7.0
Smoothing lowers Kinect MPJVE but changes MPJPE by less than 2 mm	Supported MPJVE 11.5 → 6.3, MPJPE −1.2 mm

What it means

- The VideoPose3D design matters for us. Most of our smoothness comes from the temporal window: our jitter (5.1) is close to Vicon's (4.0), while Kinect's raw jitter is 3.7 times higher.
- Kinect's lower MPJPE should not be read as "Kinect is more accurate". Our KERAAL ground truth follows the Kinect joint definition (a known limit, see section 6).

- A light smoothing filter is a free improvement before any feedback step. It removes almost all shakiness without moving the joints.

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4. E2 — Depth, camera and exercise (REHAB24-6)

Question: is depth the hardest part for us, as the REHAB24-6 paper says? Does accuracy change with camera view and exercise?

KERAAL G3: error along each camera axis (mm)

Source	x (left–right)	y (up–down)	z (depth)
Ours	23.2	25.6	**38.3**
Kinect	22.7	30.8	23.1

Exercise	Ours MPJPE	Ours depth error	Kinect MPJPE	Kinect depth error
CTK	58.5	38.6	44.5	21.8
ELK	57.8	32.4	65.5	22.7
RTK (torso rotation)	62.6	**44.0**	46.7	24.9

REHAB24-6 test (persons 1 and 2, 36 views): by camera and exercise (mm)

Group	MPJPE	x	y	z (depth)
Camera c17 (landscape)	56.0	17.8	29.0	35.6
Camera c18 (portrait)	64.6	27.1	34.7	34.1
Ex1	52.5	20.1	30.9	26.7
Ex2	55.5	21.7	31.3	29.8
Ex3	69.8	26.0	33.6	42.9
Ex4	53.7	18.7	24.9	35.0
Ex5	**111.5**	35.5	53.8	**72.8**
Ex6	59.7	23.0	30.6	35.5

Exercise names are kept as the dataset's Ex1–Ex6 codes. Check the exercise names in the REHAB24-6 readme before putting them on slides.

Prediction	Result
Depth (z) error is larger than x and y error for both sources	Partly True for ours on KERAAL (38 vs 23–26 mm). **False for Kinect**, whose depth error is its smallest (23 mm). On REHAB24 c18, depth and up–down are equal.
c17 and c18 MPJPE differ by less than 15 mm	Supported 8.6 mm (56.0 vs 64.6)

What it means

- The REHAB24-6 paper's claim holds for video-based 3D. It is exactly where a depth camera has the advantage, and the main reason Kinect still wins on position in CTK and RTK.
- RTK is a rotation toward and away from the camera. It has our largest depth error, which explains why torso rotation was the hardest exercise in every earlier report.
- Changing the camera view (landscape vs portrait) costs about 9 mm. Changing the exercise matters much more: one REHAB24 exercise (Ex5) doubles the error to 112 mm. Exercise coverage in training is a bigger risk than camera view.

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5. E3 — DTW error detection and localisation (3D pose-based feedback)

Question: can we flag a wrong repetition, and point at the wrong joint, the way EC3D does? We compare each recording with correct ones only, with no error labels used in training.

Method

- Each recording becomes 8 joint-angle curves (elbows, knees, shoulders, trunk twist and trunk bend), resampled to 60 points.
- Each recording is compared by multivariate DTW with the *correct* recordings of the other two participants. Its score is the mean of the 3 smallest distances.
- **AUC** measures how well that score separates error recordings from correct ones (0.5 = chance).
- **Localisation** is the angle with the largest single-angle DTW distance. We check whether each source picks the same angle as Vicon.
- The same 539 recordings are used for every source.

Exercise	Vicon points	Kinect	Ours
CTK	0.62 [0.51–0.73]	0.76 [0.67–0.85]	**0.73** [0.63–0.84]
ELK	0.51 [0.42–0.60]	0.52 [0.42–0.61]	0.52 [0.43–0.62]
RTK	0.64 [0.53–0.74]	0.58 [0.49–0.66]	**0.70** [0.61–0.79]

AUC with a 95 % bootstrap interval (1000 resamples of recordings). Intervals overlap for every pair of sources.

Localisation: the same "most wrong angle" as Vicon	CTK	ELK	RTK
Kinect	36 %	47 %	43 %
Ours	35 %	24 %	22 %
Chance (1 of 8 angles)		12.5 %	

Prediction	Result
Our AUC is within 0.05 of Vicon on CTK and ELK	Partly ELK: yes (0.52 vs 0.51). CTK: ours is 0.11 *higher* than Vicon, so outside the band, in the better direction.
RTK AUC is lower than CTK and ELK for every source	Not supported RTK is above ELK for every source, and ours is highest on RTK.
Localisation agreement is above chance	Supported 22–35 % for ours, 36–47 % for Kinect (chance 12.5 %)

What it means

- Our video 3D is as useful as Kinect for flagging wrong CTK and RTK repetitions with this simple, training-free method. The intervals overlap, so we cannot say one source is better.
- Vicon-point angles are weaker than ours here. This matches the earlier finding that Vicon target offsets differ per participant.
- ELK fails for every source, so the failure is in the method or the labels, not in our 3D. ELK errors may be small or affect timing more than shape, which DTW is designed to ignore.
- Training-free DTW (0.52–0.73) is clearly below the supervised random forest from RQ4 (AUC 0.85 for ours). For detection, a trained classifier is still the better tool. DTW is useful for showing *where* a movement differs from a reference.
- **Joint-level feedback is not ready.** Our "most wrong angle" agrees with Vicon only 22–35 % of the time. It is below Kinect on ELK and RTK, where our depth error is highest (E2). Telling a patient "your left knee is wrong" would often name the wrong joint.

6. Limits

- Only 3 KERAAL G3 participants, all healthy people acting errors. Results may not hold for patients.
- KERAAL's 3D ground truth follows the Kinect joint definition, which favours Kinect on position error (E1, E2).
- The "no context" condition in E1 is a test-time ablation of a model trained with context. A model trained on single frames would do better than this.
- E3 localisation is checked against Vicon's choice, not against a therapist's label of the wrong body part.
- REHAB24-6 results come from 2 test persons (36 views).
- Research 1 (UCO Physical Rehabilitation) was not tested; the dataset must be requested from the authors.
- KERAAL is CC BY-NC-SA. This page holds aggregate numbers only; per-frame outputs stay on the E: drive.

7. Next steps suggested by these results

1. Add a smoothing step to the pipeline

A 0.3 s Hamming filter after lifting. It costs nothing and cuts jitter by 76 % (5.1 → 1.2) with no loss of accuracy.

2. Attack depth on rotation exercises

RTK has the largest depth error (44 mm). Options: more rotation examples in training, or VideoPose3D's back-projection training on 1A videos (no Vicon needed).

3. Feedback at exercise level first

Use the supervised classifier (AUC 0.85) for "this repetition looks wrong". Use DTW to show the timing of the difference against a correct reference. Wait before naming a single joint.

4. Request UCO Physical Rehabilitation

Five views and 16 exercises would test the camera and exercise effects of E2 on many more people. Ex5 shows that exercise coverage matters most.

Files

- Code (with the pre-registered predictions in the header): `members/sonny/mocap3d/paper_followup/followup.py`
- Aggregate results: `members/sonny/mocap3d/paper_followup/followup_results.json`, log `followup.log`
- Model: `E:\KERAAL_mocap3d\km\kin_w_r24\run_*_s0\best.pt`
- Paper review: `Papers on the lifting models.pdf`, `Lifting_Models_for_Our_Project.docx` (class folder)

References: Pavllo et al. (2019) CVPR, arXiv:1811.11742 · REHAB24-6, Information Systems 134, Zenodo 13305826 · Zhao et al., 3D Pose Based Feedback for Physical Exercises (EC3D) · UCO Physical Rehabilitation (2023), CVIU, doi:10.1016/j.cviu.2023.103741 · Nguyen et al. (2024) KERAAL, arXiv:2407.00521